# GATE CS — Operating System

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

| Topic | Past questions | Total marks | New practice Qs |
|---|---|---|---|
| Memory management, paging, virtual memory | 21 | 37 | 2 |
| File system & disk | 12 | 22 | 2 |
| CPU scheduling | 13 | 20 | 2 |
| Synchronization (semaphores) | 11 | 20 | 2 |
| Processes, threads, system calls, interrupts | 14 | 16 | 2 |
| Deadlock | 8 | 12 | 2 |
| **Total** | **79** | **127** | **12** |

## How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

# Memory management, paging, virtual memory

Page-table sizes, TLB/EAT, page replacement, allocation. · 21 questions, 37 marks over 15 papers (≈2.5 marks/paper)

## A. Past-paper questions (21)

**2026 CS2 · Q.54 · 2 marks**

A system has a Translation Lookaside Buffer (TLB) that has a reach of 1 MB. TLB reach is defined as the total amount of physical memory that can be accessed through the TLB entries. The paging system uses pages of size 4 KB. The virtual address space is 64 GB and physical address space is 1 GB. If each TLB entry stores a 4-bit process id, page number, frame number, and a 2-bit control field, then the size of the TLB (in bytes) is _____. (answer in integer) Note: 1K=210, 1M=220, 1G=230

**2026 CS2 · Q.55 · 2 marks**

Consider contiguous allocation of physical memory to processes using variable partitioning scheme. Suppose there are 8 holes in the memory of sizes 20 KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB. Assume that no two holes are adjacent. Two processes P1 of size 16 KB and P2 of size 9 KB arrive in that order, and they are allocated memory using the best-fit technique. After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____. (answer in integer) Note: 1K=210

**2025 CS1 · Q.14 · 1 mark**

Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?
(A) 221
(B) 220
(C) 222
(D) 224

**2025 CS1 · Q.54 · 2 marks**

In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page. A process accesses the pages in the following order of page numbers: 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4. If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____ . (Answer in integer)

**2025 CS2 · Q.47 · 2 marks**
Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right): initially after - 1* 2* 3* 4* 5* 4 1 6* 4 5 1* 3* 2* - 1 1 1 1 1 1 1 6 6 6 6 6 2 - - 2 2 4 4 4 4 4 4 4 1 1 1 - - - 3 3 5 5 5 5 5 5 5 3 3 The *-marked references cause page replacements. Which one or more of the following could be the page replacement policy/policies in use?
(A) Least Recently Used page replacement policy
(B) Least Frequently Used page replacement policy
(C) Most Frequently Used page replacement policy
(D) Optimal page replacement policy

**2025 CS2 · Q.58 · 2 marks**
A computer system supports a logical address space of 232 bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is _____ . (Answer in integer)

**2024 CS1 · Q.62 · 2 marks**
Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is _____

**2024 CS2 · Q.24 · 1 mark**
Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?
(A) Allocate a new page table for a newly created process
(B) Translate a virtual address to a physical address using the page table
(C) Raise a trap when a virtual address is not found in the page table Raise a trap when a process tries to write to a page marked with read-only
(D) permission in the page table

**2024 CS2 · Q.64 · 2 marks**
Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume 1 KB = 210 bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry. An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process. The value of X+Y is _____

**2023 · Q.57 · 2 marks**
Consider the following two-dimensional array D in the C programming language, which is stored in row-major order: int D[128][128]; Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet: for (int i = 0; i < 128; i++) for (int j = 0; j < 128; j++) D[j][i] *= 10; The number of page faults generated during the execution of this code snippet is . CS

**2023 · Q.58 · 2 marks**
Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes. The value of L is . CS

**2022 · Q.38 · 2 marks**
Which one of the following statements is FALSE?
(A) The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
(B) If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
(C) The memory access time using a given inverted page table is always same for all incoming virtual addresses.
(D) In a system that uses hashed page tables, if two distinct virtual addresses V1 and V2 map to the same value while hashing, then the memory access time of these addresses will not be the same.

**2022 · Q.64 · 2 marks**
Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5,3, 4, 6, 7, 7,1,5, 6,1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is_____.

**2021 · Q.11 · 1 mark** [OCR]
In the context of operating systems, which of the following statements is/are correct with respect to paging?
(A) | Paging helps solve the issue of external fragmentation.
(B) | Page size has no impact on internal fragmentation.
(C) | Paging incurs memory overheads.
(D) | Multi-level paging is necessary to support pages of different sizes.

**2020 · Q.11 · 1 mark** [OCR]
Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process’s memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?
(A) The hole created by first fit is always larger than the hole created by next fit. (8) The hole created by worst fit is always larger than the hole created by first fit. (©) The hole created by best fit is never larger than the hole created by first fit. (0) The hole created by next fit is never larger than the hole created by best fit.

**2020 · Q.53 · 2 marks** [OCR]
Consider a paging system that uses 1I-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%, Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is

**2019 · Q.33 · 2 marks** [OCR]
Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressible. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?
(A) 16x2!°
(B) 256x2!°
(C) 4x20
(D) 8x27

**2018 · Q.10 · 1 mark**
Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units. Which one of the following is the correct expression for the page fault rate experienced by the process?
(A) (D – M) / (X – M)
(B) (X – M) / (D – M)
(C) (D – X) / (D – M)
(D) (X – M) / (D – X)

**2017 CS1 · Q.40 · 2 marks** [OCR]
Recall that Belady’s anomaly is that the page-fault rate may increase as the number of allocated frames increases. Now. consider the following statements: S1: Random page replacement algorithm (where a page chosen at random is replaced) suffers from Belady’s anomaly $2: LRU page replacement algorithm suffers from Belady’s anomaly Which of the following is CORRECT?
(A) S1 is true. S2 is true
(B) S1 is true, S2 is false
(C) S1 is false. S2 is true
(D) S1 is false. S2 is false

**2016 · Q.47 · 2 marks**
Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is megabytes.

**2016 · Q.49 · 2 marks**
Consider a computer system with ten physical page frames. The system is provided with an access sequence (a1 , a2 , . . . , a20 , a1 , a2 , . . . , a20 ), where each ai is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is .

## B. New practice questions with solutions

**1.** (1 mark · NAT) A system has 48-bit virtual addresses, 8 KB pages and 8-byte page table entries, and uses a single-level page table. The size of the page table in GB is _____.

**256.** The page offset takes 13 bits, leaving $2^{35}$ pages. $2^{35} \times 8$ bytes = $2^{38}$ bytes = 256 GB.

**2.** (2 marks · NAT) A process with 3 frames references pages 2 3 2 1 5 2 4 5 3 2 5 2. Let F be the number of page faults under FIFO and L the number under LRU, with all frames initially empty. F − L is _____.

**2.** FIFO gives 9 faults and LRU gives 7 (optimal replacement would give 6).

# File system & disk

Disk scheduling, disk access time, file allocation, inodes. · 12 questions, 22 marks over 15 papers (≈1.5 marks/paper)

## A. Past-paper questions (12)

**2026 CS1 · Q.59 · 2 marks**
Consider a hard disk with a rotational speed of 15000 rpm. The time to move the read/write head from a track to its adjacent track is 1 millisecond. Initially, the head is on track 0. The number of sectors per track is 400. The sector size is 1024 bytes. It is necessary to transfer data from 10 randomly located sectors in each of the following tracks in the order: 5, 12 and 7. The total time for the data transfer (in milliseconds) from the hard disk is _____. (rounded off to one decimal place)

**2026 CS2 · Q.53 · 2 marks**
To keep track of free blocks in a file system, one of the two approaches is generally used – using bitmaps (bit vectors) or using linked lists. Consider that the linked list approach is used to keep track of free blocks in a file system. Assume that the disk size is 16 GB, block size is 2 KB, and block numbers used are 32-bit long. A single pointer of size 4 bytes is used in each block of the list to point to the next block of the list. The number of blocks required to hold the free disk block numbers is _____. (answer in integer) Note: 1K=210 and 1G=230

**2025 CS1 · Q.51 · 2 marks**
A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, 1K = 210 and 1M = 220 . The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)

**2024 CS1 · Q.54 · 2 marks**
Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____

**2024 CS2 · Q.53 · 2 marks**
Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____

**2022 · Q.63 · 2 marks**
Consider two files systems A and B , that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50 th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are nA and nB , respectively, then the value of nA  nB is_____.

**2021 · Q.15 · 1 mark** [OCR]
Consider a linear list based directory implementation in a file system. Each direc- ) P \ tory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory foo. Which of the following operations will necessarily require a full scan of foo for successful completion?
(A) | Creation of a new file in foo
(B) | Deletion of an existing file from foo
(C) | Renaming of an existing file in foo
(D) | Opening of an existing file in foo CS, Set-1 - Copyright © GATE 2021

**2020 · Q.35 · 2 marks** [OCR]
Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time. (P, 155), (Q, 85), (R, 110), (S, 30), (T, 115) Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests. Which one of the following statements is FALSE?
(A) T is serviced before P. (8) Qis serviced after S, but before T. (©) The head reverses its direction of movement between servicing of Q and P. (0) R is serviced before P.

**2019 · Q.42 · 2 marks** [OCR]
The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) GB. cs 10/15 GATE 2019 Computer Science and Information Technology Set-2

**2018 · Q.53 · 2 marks**
Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time: [120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1] Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

**2017 CS2 · Q.8 · 1 mark** [OCR]
In a file allocation system, which of the following allocation scheme(s) can be used if no extemal fragmentation is allowed? I. Contiguous Il. Linked Ill. Indexed
(A) Land III only
(B) Il only
(C) II only
(D) I and III only

**2016 · Q.48 · 2 marks**
Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is .

## B. New practice questions with solutions

**1.** (1 mark · NAT) The disk head is at cylinder 40. The queue holds requests for cylinders 12, 90, 55, 30, 70, 5, 100. Under SSTF scheduling the total head movement is _____ cylinders.

**130.** The service order is 30, 12, 5, 55, 70, 90, 100, with movements 10 + 18 + 7 + 50 + 15 + 20 + 10 = 130.

**2.** (2 marks · NAT) A Unix-like inode has 10 direct pointers, 1 single-indirect pointer and 1 double-indirect pointer. The block size is 2 KB and each pointer is 4 bytes. The maximum file size in KB is _____.

**525332.** Each indirect block holds 512 pointers. The file can have $10 + 512 + 512^2 = 262{,}666$ blocks, and $262{,}666 \times 2$ KB = 525,332 KB.

# CPU scheduling

FCFS/SJF/SRTF/RR/priority: waiting and turnaround time, starvation. · 13 questions, 20 marks over 15 papers (≈1.3 marks/paper)

## A. Past-paper questions (13)

**2026 CS1 · Q.64 · 2 marks**

Consider a CPU that has to execute two types of processes. The first type, Actuators (A), requires a CPU burst of 6 seconds. The second type, Controllers (C), requires a CPU burst of 8 seconds. A new process of type A arrives at time t = 10, 20, 30, 40, and 50 (in seconds). Similarly, a new process of type C arrives at time t = 11, 22, 33, 44, and 55 (in seconds). The CPU scheduling policy is First Come First Serve (FCFS). The first process of type A starts running at t = 10 seconds. The average waiting time (in seconds) for the 10 processes is _____. (rounded off to one decimal place)

**2026 CS2 · Q.23 · 1 mark**

Which one of the following CPU scheduling algorithms cannot be preemptive?
(A) Shortest Remaining Time First (SRTF) Scheduling
(B) First Come First Serve (FCFS) Scheduling
(C) Round Robin Scheduling
(D) Priority Scheduling

**2025 CS1 · Q.38 · 2 marks**

A computer has two processors, M1 and M2 . Four processes P1 , P2 , P3 , P4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows: • M1 uses priority of execution for the processes as, P1 > P3 > P2 > P4 , i.e., P1 and P4 have highest and lowest priorities, respectively. • M2 uses priority of execution for the processes as, P2 > P3 > P4 > P1 , i.e., P2 and P1 have highest and lowest priorities, respectively. A process Pi is scheduled to a processor Mk , if the processor is free and no other process Pj is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?
(A) 9.00
(B) 8.75
(C) 6.50
(D) 7.50

**2025 CS2 · Q.26 · 1 mark**

Processes P1, P2, P3, P4 arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times. Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?
(A) 22
(B) 15
(C) 37
(D) 19

**2024 CS2 · Q.37 · 2 marks**
Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time. A (0, 10), B (2, 6), C (4, 3), and D (6, 7). Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?
(A) SRTF = 6, NP-SJF = 7
(B) SRTF = 6, NP-SJF = 7.5
(C) SRTF = 7, NP-SJF = 7.5
(D) SRTF = 7, NP-SJF = 8.5 Which one of the following CIDR prefixes exactly represents the range of

**2023 · Q.27 · 1 mark**
Which one or more of the following CPU scheduling algorithms can potentially cause starvation?
(A) First-in First-Out
(B) Round Robin
(C) Priority Scheduling
(D) Shortest Job First CS

**2022 · Q.42 · 2 marks**
Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time t = 0. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?
(A) P = 4, Q = 10, R = 6, S = 2
(B) P = 2, Q = 9, R = 5, S = 1
(C) P = 4, Q = 12, R = 5, S = 4
(D) P = 3, Q = 7, R = 7, S = 3

**2021 · Q.25 · 1 mark** [OCR]
Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the mini- mum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to nearest integer) is _____ milliseconds. CS, Set-1 - Copyright © GATE 2021

**2020 · Q.50 · 2 marks** [OCR]
Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order FP PP, Processes Be | ky | Be Ae Burst time (inms) | 8 | 7 | 2 | 4 If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is

**2019 · Q.41 · 2 marks** [OCR]
Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below: Process Pl P2 P3 P4 Arrival time 0 1 3 4 CPU burst time 3 1 3 Z These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is

**2017 CS1 · Q.24 · 1 mark** [OCR]
Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) as given below : Process | Arrival time | Burst time Pl 0 F P2 3 3 P3 5 2] P4 6 2 If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes. then the average waiting time across all processes is milliseconds. =====PAGE 13=====

**2017 CS2 · Q.51 · 2 marks** [OCR]
Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds). and priority (0 is the highest priority) shown below. None of the processes have I/O burst time. Process Arrival Time | Burst Time Priority P, 0 in 2 P5 5 28 0 Be 12 2 3 Py 2 10 1 Ps 9 16 4 The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is

**2016 · Q.20 · 1 mark**
Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?
(A) Shortest remaining time first
(B) Round-robin with time quantum less than the shortest CPU burst
(C) Uniform random
(D) Highest priority first with priority proportional to CPU burst length

## B. New practice questions with solutions

**1.** (1 mark · NAT) Processes P1, P2, P3 arrive at times 0, 1, 2 with CPU bursts 6, 4, 2. Under FCFS the average waiting time, to two decimal places, is _____.

**4.33.** The waiting times are 0, 5 and 8, and 13/3 = 4.33.

**2.** (2 marks · NAT) Four processes arrive at times 0, 1, 2, 3 with bursts 8, 4, 2, 1. Under preemptive SJF (SRTF) the average turnaround time is _____.

**6.5.** The completion times are P1 = 15, P2 = 8, P3 = 4 and P4 = 5. The turnaround times are 15, 7, 2 and 2, which sum to 26 and average 6.5.

# Synchronization (semaphores)

Semaphore solutions, race conditions, mutual exclusion. · 11 questions, 20 marks over 15 papers (≈1.3 marks/paper)

## A. Past-paper questions (11)

**2026 CS2 · Q.51 · 2 marks**

Consider three processes P1, P2, and P3 running identical code, as shown in the pseudocode below. A and B are two binary semaphores initialized to 1 and 0, respectively. X is a shared variable initialized to 0. Each line in the pseudocode is executed atomically. Pseudocode of P1, P2, and P3 Wait(A); Print(*); X = X+1; If (X == 2) { Print($); Signal(B); } Signal(A); Wait(B); Print(#); Signal(B); Assume that any of the three processes can start to execute first and context switching can happen between these processes at any arbitrary time and in any arbitrary order. Which of the following patterns is/are possible to be generated as an outcome of the execution of these three processes?

(A) **$*###

(B) **$#*##

(C) **$###*#

(D) ***$###

**2024 CS1 · Q.40 · 2 marks**

Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially a = b = 1. Though context switching between threads can happen at any time, each statement of T1 or T2 is executed atomically without interruption. T1 T2 a = a + 1; b = 2 * b; b = b + 1; a = 2 * a; Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution?

(A) (a = 4, b = 4); (a = 3, b = 3); (a = 4, b = 3)

(B) (a = 3, b = 4); (a = 4, b = 3); (a = 3, b = 3)

(C) (a = 4, b = 4); (a = 4, b = 3); (a = 3, b = 4)

(D) (a = 2, b = 2); (a = 2, b = 3); (a = 3, b = 4)

**2024 CS2 · Q.46 · 2 marks**

two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below. // code of T1 // code of T2 wait(s1); wait(s1); x = x+1; x = x+1; print(x); print(x); wait(s2); signal(s2); signal(s1); signal(s1); Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

(A) T1 runs first and prints 1, T2 runs next and prints 2

(B) T2 runs first and prints 1, T1 runs next and prints 2

(C) T1 runs first and prints 1, T2 does not print anything (deadlock)

(D) T2 runs first and prints 1, T1 does not print anything (deadlock) Let A be an n × n matrix over the set of all real numbers R. Let B be a matrix

**2023 · Q.38 · 2 marks**

Consider the two functions incr and decr shown below. incr(){ decr(){ wait(s); wait(s); X = X+1; X = X-1; signal(s); signal(s); } } There are 5 threads each invoking incr once, and 3 threads each invoking decr once, on the same shared variable X. The initial value of X is 10. Suppose there are two implementations of the semaphore s, as follows: I-1: s is a binary semaphore initialized to 1. I-2: s is a counting semaphore initialized to 2. Let V1, V2 be the values of X at the end of execution of all the threads with implementations I-1, I-2, respectively. Which one of the following choices corresponds to the minimum possible values of V1, V2, respectively?

(A) 15, 7

(B) 7, 7

(C) 12, 7

(D) 12, 8 CS

**2022 · Q.19 · 1 mark**

Consider the following threads, T1, T2, and T3 executing on a single processor, synchronized using three binary semaphore variables, S1, S2, and S3, operated upon using standard wait() and signal(). The threads can be context switched in any order and at any time. T1 T2 T3 while(true){ while(true){ while(true){ wait(S3); wait(S1); wait(S2); print(“C”); print(“B”); print(“A”); signal(S2); } signal(S3); } signal(S1); } Which initialization of the semaphores would print the sequence BCABCABCA....?

(A) S1 = 1; S2 = 1; S3 = 1

(B) S1 = 1; S2 = 1; S3 = 0

(C) S1 = 1; S2 = 0; S3 = 0

(D) S1 = 0; S2 = 1; S3 = 1

**2021 · Q.46 · 2 marks** [OCR]

Consider the following pseudocode, where S is a semaphore initialized to 5 in line#2 and counter is a shared variable initialized to 0 in line#1. Assume that the incre- ment operation in line#7 is not atomic. 1. int counter = 0; 2. Semaphore S = init(5); 3. void parop(void) 4.{ 5. wait (S); 6. wait(S); 7. counter++; 8. signal (S) ; 9. signal (S) ; 10.} If five threads execute the function parop concurrently, which of the following program behavior(s) is/are possible?

(A) | The value of counter is 5 after all the threads successfully complete the execution of parop.

(B) | The value of counter is 1 after all the threads successfully complete the execution of parop.

(C) | The value of counter is 0 after all the threads successfully complete the execution of parop.

(D) | There is a deadlock involving all the threads. CS, Set-1 - Copyright © GATE 2021

**2020 · Q.34 · 2 marks** [OCR]
Each of a set of n processes executes the following code using two semaphores a and b initialized to 1 and 0, respectively. Assume that count is a shared variable initialized to 0 and not used in CODE SECTION P. CODE SECTION P wait(a); count=count+1; if (count==n) signal (b); signal(a); wait(b); signal(b); CODE SECTION Q What does the code achieve?
(A) It ensures that no process executes CODE SECTION Q before every process has finished CODE SECTION P. (8) It ensures that at most two processes are in CODE SECTION Q at any time. (©) It ensures that all processes execute CODE SECTION P mutually exclusively. (0) It ensures that at most n—1 processes are in CODE SECTION P at any time.

**2019 · Q.23 · 1 mark** [OCR]
Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D that has been initialized to 100. Pl P2 P3 D=D+20 D=D-50 D=D+10 The processes are executed on a uniprocessor system running a time-shared operating system. If the minimum and maximum possible values of D after the three processes have completed execution are X and Y respectively, then the value of Y — X is

**2018 · Q.40 · 2 marks**
Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N. Three semaphores empty, full and mutex are defined with respective initial values of 0, N and 1. Semaphore empty denotes the number of available slots in the buffer, for the consumer to read from. Semaphore full denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P, Q, R, and S, in the code below can be assigned either empty or full. The valid semaphore operations are: wait() and signal(). Producer: Consumer: do{ do{ wait(P); wait(R); wait(mutex); wait(mutex); //Add item to buffer //Consume item from buffer signal(mutex); signal(mutex); signal(Q); signal(S); }while(1); }while(1); Which one of the following assignments to P, Q, R and S will yield the correct solution?
(A) P: full, Q: full, R: empty, S: empty
(B) P: empty, Q: empty, R: full, S: full
(C) P: full, Q: empty, R: empty, S: full
(D) P: empty, Q: full, R: full, S: empty

**2017 CS1 · Q.27 · 2 marks** [OCR]
A multithreaded program P executes with x number of threads and uses y number of locks for ensuring mutual exclusion while operating on shared memory locations. All locks in the program are non- reentrant. i.e., if a thread holds a lock /. then it cannot re-acquire lock / without releasing it. If a thread is unable to acquire a lock. it blocks until the lock becomes available. The minimum value of x and the minimum value of v together for which execution of P can result in a deadlock are:
(A) x=Ly=2 (@) x=2,y=1
(C) x=2,y=2
(D) x=Ly=1 =====PAGE 14=====

**2016 · Q.50 · 2 marks**
Consider the following proposed solution for the critical section problem. There are n processes: P0 . . . Pn−1 . In the code, function pmax returns an integer not smaller than any of its arguments. For all i, t[i] is initialized to zero. Code for Pi : do { c[i]=1; t[i] = pmax(t[0],. . .,t[n-1])+1; c[i]=0; for every j 6= i in {0,. . .,n-1} { while (c[j]); while (t[j] != 0 && t[j]<=t[i]); } Critical Section; t[i]=0; Remainder Section; } while (true); Which one of the following is TRUE about the above solution?
(A) At most one process can be in the critical section at any time
(B) The bounded wait condition is satisfied
(C) The progress condition is satisfied
(D) It cannot cause a deadlock

## B. New practice questions with solutions

**1.** (1 mark · MCQ) Semaphores s1 = 1 and s2 = 0. Process P1 loops forever: P(s1); print X; V(s2). Process P2 loops forever: P(s2); print Y; V(s1). The output is

A. XYXYXY… (B) XXYYXXYY… (C) deadlock with no output (D) YXYXYX…

**(A).** The two semaphores force strict alternation, starting with P1.

**2.** (2 marks · MCQ) A bounded-buffer producer-consumer solution uses semaphores empty = N, full = 0 and mutex = 1. A programmer swaps the first two lines of the producer to `P(mutex); P(empty);`, leaving the consumer unchanged. Which statement is TRUE?

A. The solution is still correct.
B. Mutual exclusion on the buffer is violated.
C. A deadlock can occur when the buffer is full.
D. A deadlock can occur when the buffer is empty.

**(C).** When the buffer is full the producer holds mutex and blocks on empty. The consumer needs mutex to remove an item, so neither can proceed.

# Processes, threads, system calls, interrupts

Context switch, threads vs processes, process states, fork(), system calls. · 14 questions, 16 marks over 15 papers (≈1.1 marks/paper)

## A. Past-paper questions (14)

**2026 CS1 · Q.63 · 2 marks**
Consider the following program snippet. Assume that the program compiles and runs successfully. Further, assume that the fork() system call is always successful in creating a process. int main () { int i; for (i = 0; i < 3; i++){ if (fork() == 0){ continue; } break; } printf("Hello!"); return 0; } The total number of times that the printf statement gets executed is _____. (answer in integer)

**2025 CS1 · Q.29 · 1 mark**
Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. (Answer in integer) int main() { int x=0,i=0; scanf("%d",&x); for(i=0; i<20; i++) { x = x+20; printf("%d\n",x); } return 0; }

**2024 CS1 · Q.24 · 1 mark**
Which of the following statements about threads is/are TRUE?
(A) Threads can only be implemented in kernel space
(B) Each thread has its own file descriptor table for open files
(C) All the threads belonging to a process share a common stack
(D) Threads belonging to a process are by default not protected from each other

**2024 CS1 · Q.25 · 1 mark**
Which of the following process state transitions is/are NOT possible?
(A) Running to Ready
(B) Waiting to Running
(C) Ready to Waiting
(D) Running to Terminated

**2024 CS1 · Q.57 · 2 marks**
Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors. int x = 3; while(x > 0) { fork(); printf("hello"); wait(NULL); x--; } The total number of times the printf statement is executed is _____

**2024 CS2 · Q.25 · 1 mark**
Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?
(A) P makes a blocking system call to read a block of data from the disk
(B) P tries to access a page that is in the swap space, triggering a page fault
(C) An interrupt is raised by the disk to deliver data requested by some other process
(D) A timer interrupt is raised by the hardware

**2023 · Q.22 · 1 mark**
Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process?
(A) Page table base register
(B) Stack pointer
(C) Program counter
(D) General purpose registers CS

**2023 · Q.23 · 1 mark**
Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?
(A) Function Call
(B) malloc Call
(C) Page Fault
(D) System Call CS

**2021 · Q.14 · 1 mark** [OCR]
Which of the following standard C library functions will always invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?
(A) | exit
(B) | malloc
(C) | sleep
(D) | strlen CS, Set-1 - Copyright © GATE 2021

**2020 · Q.12 · 1 mark** [OCR]
Consider the following statements about process state transitions for a system using preemptive scheduling. I. A running process can move to ready state. II. A ready process can move to running state. III. A blocked process can move to running state. IV. A blocked process can move to ready state. Which of the above statements are TRUE? (a) I, IL, and III only (8) Il and III only (©) I, IL, and IV only (0) I, Il, IIL, and IV

**2019 · Q.17 · 1 mark** [OCR]
The following C program is executed on a Unix/Linux system: #include <unistd.h> int main() { int i; for (i=0; i<10; i++) if (i%2 == 0) fork(); return 0; } The total number of child processes created is

**2018 · Q.9 · 1 mark**
The following are some events that occur after a device controller issues an interrupt while process L is under execution. (P) The processor pushes the process status of L onto the control stack. (Q) The processor finishes the execution of the current instruction. (R) The processor executes the interrupt service routine. (S) The processor pops the process status of L from the control stack. (T) The processor loads the new PC value based on the interrupt. Which one of the following is the correct order in which the events above occur?
(A) QPTRS
(B) PTRSQ
(C) TRPQS
(D) QTPRS

**2017 CS1 · Q.18 · 1 mark** [OCR]
Threads of a process share
(A) global variables but not heap.
(B) heap but not global variables.
(C) neither global variables nor heap.
(D) both heap and global variables.

**2017 CS2 · Q.7 · 1 mark** [OCR]
Which of the following is/are shared by all the threads in a process? I. Program counter II. Stack Ill. Address space IV. Registers
(A) Land II only
(B) If only
(C) IV only
(D) III and IV only

## B. New practice questions with solutions

**1.** (1 mark · NAT) The program below is run. The number of X lines printed is _____.

```
int main() {
    if (fork()) fork();
    fork();
    printf("X\n");
}
```

**6.** After the first statement there are 3 processes, because the child skips the inner fork. The next `fork()` doubles them to 6, and each prints once.

**2.** (2 marks · MSQ) Which statements are TRUE?

A. Threads of one process share the heap and global data.
B. Each thread has its own stack and register set.
C. A context switch between two threads of the same process requires switching the page table.
D. In a many-to-one thread model, a blocking system call by one user thread blocks the whole process.

**(A), (B), (D).** Threads of the same process share one address space, so (C) is false.

# Deadlock

Banker's algorithm, deadlock conditions, resource counts. · 8 questions, 12 marks over 15 papers (≈0.8 marks/paper)

## A. Past-paper questions (8)

**2026 CS1 · Q.29 · 1 mark**

With respect to deadlocks in an operating system, which of the following statements is/are FALSE?
(A) Banker’s algorithm is used to prevent deadlocks
(B) Deadlock formation can be prevented by ensuring that the hold and wait condition is not allowed
(C) An assignment edge in a resource allocation graph is marked from a process to a resource
(D) A safe state guarantees that all processes can finish without formation of a deadlock

**2026 CS1 · Q.35 · 1 mark**

Consider a system consisting of k instances of a resource R, being shared by 5 processes. Assume that each process requires a maximum of two instances of resource R and a process can request or release only one instance at a time. Further, a process can request the second instance of the resource only after acquiring the first instance. The minimum value of k for the system to be deadlock-free is _____. (answer in integer)

**2025 CS2 · Q.48 · 2 marks**

P = {P1 , P2 , P3 , P4 } consists of all active processes in an operating system. R = {R1 , R2 , R3 , R4 } consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: R1 → P1 , R2 → P2 , R3 → P3 , R4 → P4 (the assignment edge R1 → P1 means resource R1 is assigned to process P1 , and so on for others) Claim edges: P1 → R2 , P2 → R3 , P3 → R1 , P2 → R4 , P4 → R2 (the claim edge P1 → R2 means process P1 is waiting for resource R2 , and so on for others) Which of the following statement(s) is/are CORRECT?
(A) Aborting P1 makes the system deadlock free.
(B) Aborting P3 makes the system deadlock free.
(C) Aborting P2 makes the system deadlock free.
(D) Aborting P1 and P4 makes the system deadlock free.

**2022 · Q.26 · 1 mark**

Which of the following statements is/are TRUE with respect to deadlocks?
(A) Circular wait is a necessary condition for the formation of deadlock.
(B) In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock.
(C) If the current allocation of resources to processes leads the system to unsafe state, then deadlock will necessarily occur.
(D) In the resource-allocation graph of a system, if every edge is an assignment edge, then the system is not in deadlock state.

**2019 · Q.39 · 2 marks** [OCR]
Consider the following snapshot of a system running n concurrent processes. Process i is holding X; instances of a resource R, 1 < i < n. Assume that all instances of R are currently in use. Further, for all i, process i can place a request for at most Y; additional instances of R while holding the X; instances it already has. Of the n processes, there are exactly two processes p and q such that ¥, = ¥Y, = 0. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?
(A) Xp + X, <Min {¥, |1<k <n,k4p,k4q}
(B) X, + Xq< Max {% |1<k<n,k+p,k+#q}
(C) Min (Xp ,Xq) = Min {%, |1 <k <n, k¥p,k¥q}
(D) Min (X,,Xq) = Max {y |1<k <n, k¥p,k¥q}

**2018 · Q.24 · 1 mark**
Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ____.

**2018 · Q.39 · 2 marks**
In 0 a system, there are three types of resources: E, F and G. Four processes P0, P1, P2 and P3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, Max[P2,F] is the maximum number of instances of F that P2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation. Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available. Allocation Max E F G E F G P0 1 0 1 P0 4 3 1 P1 1 1 2 P1 2 1 4 P2 1 0 3 P2 1 3 3 P3 2 0 0 P3 5 4 1 From the perspective of deadlock avoidance, which one of the following is true?
(A) The system is in safe state.
(B) The system is not in safe state, but would be safe if one more instance of E were available
(C) The system is not in safe state, but would be safe if one more instance of F were available
(D) The system is not in safe state, but would be safe if one more instance of G were available CS 13/20 GATE 2018

**2017 CS2 · Q.33 · 2 marks** [OCR]
A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below: Process Current Allocation Maximum Requirement PI 3 7 P2 1 6 P3 3 5 Which of the following best describes current state of the system?
(A) Safe. Deadlocked
(B) Safe, Not Deadlocked
(C) Not Safe, Deadlocked
(D) Not Safe, Not Deadlocked

## B. New practice questions with solutions

**1.** (1 mark · NAT) Five processes each need at most 3 units of a single resource type. The minimum total number of units that guarantees deadlock freedom is _____.

**11.** $n(m - 1) + 1 = 5 \times 2 + 1$.

**2.** (2 marks · MCQ) A system has 4 processes and 3 resource types with Available = (2, 1, 1).

| Process | Allocation | Max |
|---|---|---|
| P0 | 0 1 0 | 7 2 3 |
| P1 | 2 0 0 | 3 2 2 |
| P2 | 3 0 2 | 9 0 2 |
| P3 | 2 1 1 | 2 2 2 |

Which is a safe sequence?

A. P3, P1, P2, P0
B. P1, P3, P2, P0
C. P3, P2, P1, P0
D. The state is unsafe.

**(A).** The Need matrix is P0 (7,1,3), P1 (1,2,2), P2 (6,0,0), P3 (0,1,1). Only P3 can run first. Then Available = (4,2,2) and only P1 fits. Then Available = (6,2,2) and only P2 fits. Then Available = (9,2,4) and P0 fits. This is the only safe sequence.